LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Acids, Bases and Salts

Indicators

Indicator	In acid	In base
Blue litmus	turns red	no change
Red litmus	no change	turns blue
Phenolphthalein	colourless	pink
Methyl orange	red	yellow
Turmeric	yellow (no change)	reddish-brown

Olfactory indicators (onion, vanilla, clove oil): their smell cannot be detected in a basic solution.

Key reactions

- Acid + metal → salt + hydrogen: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (gas burns with a pop)
- Base + metal (some): $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$ (sodium zincate)
- Acid + carbonate/hydrogencarbonate → salt + water + CO_2: $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
- CO_2 turns lime water milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$
- Neutralisation (acid + base → salt + water): $NaOH + HCl \rightarrow NaCl + H_2O$
- Acid + metal oxide → salt + water: $CuO + 2HCl \rightarrow CuCl_2 + H_2O$ (solution turns blue-green)

pH and daily life

- pH 0–7 acidic · 7 neutral · 7–14 basic. **Lower pH = stronger acid** (more H^+ ions).
- Acids give H^+ (H_3O^+) ions only in water. Bases soluble in water are called **alkalis** and give OH^- ions.
- Tooth decay starts below pH 5.5. Rain with pH below 5.6 is acid rain.
- Bee sting (methanoic acid): apply baking soda. Indigestion: antacid such as milk of magnesia, $Mg(OH)_2$.

Important salts

Salt	Formula	How it's made / key use
Caustic soda	NaOH	Chlor-alkali process: $2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$
Bleaching powder	$CaOCl_2$	$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$ · bleaching, disinfecting water
Baking soda	$NaHCO_3$	$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$ · antacid, baking powder
Washing soda	$Na_2CO_3 \cdot 10H_2O$	Recrystallise sodium carbonate · glass, soap, removes permanent hardness
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$	Heat gypsum at 373 K · plasters for fractures, toys, decoration

Water of crystallisation: copper sulphate crystals $CuSO_4 \cdot 5H_2O$ are blue. On heating they lose water and turn white. Adding water brings back the blue colour.
Baking powder = baking soda + a mild edible acid (tartaric acid). The CO_2 released makes the cake soft and spongy.

Common mistakes

- Thinking pH 5 is more acidic than pH 3. It's the other way round.
- Adding water to concentrated acid. Always add **acid to water**, slowly and with stirring.
- Mixing up baking soda ($NaHCO_3$) and washing soda ($Na_2CO_3 \cdot 10H_2O$).

Quick check

1. Which is more acidic: a solution of pH 3 or one of pH 5?

2. Write the formula of Plaster of Paris and one use.

3. Why does dry HCl gas not change the colour of dry litmus paper?